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Methane category, immune response, feed efficiency, and rumen microbial community in lactating dairy cows

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ABSTRACT

This study aimed to assess relationships of enteric methane (CH_4) yield (g/kg of DMI) with immune response, feed efficiency (ECM/DMI), and rumen microbiome in dairy cows, both in early and in late lactation. The DMI, BW, ECM yield, and CH_4 emission were measured in respiration chambers in early (n = 20, 32 ± 7 DIM) and nonpregnant late lactating (n = 14, 359 ± 90 DIM) multiparous Holstein cows. The in vitro immune response was studied in response to (1) LPS using whole blood, and (2) phytohemagglutinin and concanavalin A using peripheral blood mononuclear cells. The DNA was extracted from rumen content samples (esophageal tubing) for 16S rRNA microbial analysis. Cows were divided retrospectively into an equal number of cows with low (LMY) and high (HMY) CH_4 yield within each lactation group. In early lactation, CH_4 yields in LMY (n = 10) and HMY cows (n = 10) were (LSM) 18.7 and 25.3 g/kg of DMI, respectively. In late lactation, CH_4 yields in LMY (n = 7) and HMY cows (n = 7) were 22.8 and 26.8 g/kg of DMI, respectively. Statistical analysis was performed separately for each lactation group. In early lactation, we found that whole blood and isolated peripheral blood mononuclear cells from LMY compared with HMY animals were less responsive to stimulants in vitro. In addition, feed conversion efficiency was lower in LMY than HMY cows, and the relative abundance of the archaeal genus *Methanosphaera* and the bacterial genus *Marvinbryantia* were higher. In late lactation, we observed no differences in immune response and feed conversion efficiency between LMY and HMY cows. Still, in LMY cows several bacterial genera including *Prevotella* 7, *Ruminococcus gauvreauii* group, and *Shuttleworthia* were enriched, whereas in HMY cows *Methanobrevibacter*, *Veillonellaceae* UCG-001, *Succinivibrionaceae* UCG-002, *Rikenellaceae* RC9, and CAG-352 were enriched. The results indicate that in early lactation the animals with low CH_4 yield reach energy balance faster, at the expense of an inadequate immune response. Meanwhile, increased CH_4 yield in early lactation may reflect higher rumen fermentation activity, fostering feed efficiency and energy availability for supporting immune function.
Key words: methane emission, energy efficiency, immune cell proliferation, rumen microbe

INTRODUCTION

In dairy cows, CH_4 emissions can vary greatly between animals, even among cows consuming similar diets and at comparable stages of lactation (Garnsworthy et al., 2012; Lyons et al., 2018; Fresco et al., 2023). Methane yield (g/kg of DMI) is a trait used to understand variation of emission from fermentation processes occurring in the rumen (Cabezas-Garcia et al., 2017). Dairy cows with a low compared with a high CH_4 yielding category were reported to have lower digestibility of fiber, but similar feed conversion efficiency (**FCE**; ECM/DMI) and similar or greater milk production efficiency (**MPE**; ECM/100 kg BW or ECM/kg $BW^{0.75}$; Denninger et al., 2020; Stepanchenko et al., 2023).

Methane yield fluctuates throughout the lactation cycle with the highest variability mostly occurring during early lactation, primarily due to differences in body condition loss and endocrine changes affecting feed intake (Bielak et al., 2016; Fresco et al., 2023). During early lactation, cows typically experience negative energy balance (**EB**) caused not only by increased metabolic activity for milk production but also by the rapid increase in energy requirements, which cannot be met by an increase in energy intake, thus challenging the immune homeostasis and function (Schwarm et al.,

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2013). Strikingly, cows vary considerably in the time postpartum required to reach positive EB (Vossebeld et al., 2022). There is a lack of evidence, but in theory, low CH_4 yielding cows may lose relatively less feed energy as CH_4, resulting in more feed energy available to partition toward the mammary gland and immune response in early lactation. However, it could also be speculated that the low CH_4 yield in early lactation is indicative of a delayed adaptation of rumen microorganisms to the overall increase in DMI during that time.

Meese et al. (2020) demonstrated that low CH_4 intensity (CH_4/ECM) cows reveal less proliferation of peripheral blood mononuclear cells (**PBMC**) compared with high CH_4 intensity cows. The authors assumed that low CH_4 intensity cows might lack sufficient digestive energy to elicit an adequate immune response (Meese et al., 2020). Therefore, exploring the association between immune response and CH_4 emissions will enhance our understanding of the variation in CH_4 yield. Still, knowledge in this area remains limited. Additionally, some studies reported how the variation in CH_4 yield during early lactation is reflected by the rumen microbial community in dairy cows, yet findings are inconsistent (Lyons et al., 2018; Ahvenjärvi et al., 2024; Marcos et al., 2024). In the study by Lyons et al. (2018), lower CH_4 yield in early lactation was associated with an elevated abundance of *Methanosphaera* and bacterial genera *Succiniclasticum*, but none of these differences were found by Ahvenjärvi et al. (2024), suggesting host-microbiome interactions, but not conclusive ones. As cows progress toward late lactation, defined ≥200 DIM, variation of CH_4 yield under similar feeding regimen and management conditions is relatively smaller compared with early lactation, likely due to a more stable feed intake and endocrine and metabolic status (Bielak et al., 2016; Fresco et al., 2023). In pregnant cows, feed intake and blood immune response was reported to be suppressed around 30 d before calving (Trevisi and Minuti, 2018). However, nonpregnant late lactating cows are expected to be in positive EB with minimum challenges for immune homeostasis (Meese et al., 2018; Moreira et al., 2021). Still, whether and to what extent the immune response differs in cows with distinct CH_4 yield remains unknown. In addition, studies reported a link between CH_4 yield and rumen microbial community in late lactating cows, but rumen microbial abundance was inconsistent (Lyons et al., 2018; Stepanchenko et al., 2023; Marcos et al., 2024). Stepanchenko et al. (2023) found only small differences in methanogenic archaeal community in cows differing in CH_4 yield in late lactation, but large differences were observed in bacterial communities involved in carbohydrate metabolism, similar to the results of Lyons et al. (2018).

Therefore, the present study aimed to assess relationships of CH_4 categories (low and high CH_4 yield) with immunological traits, feed efficiency measures, and rumen microbial community in lactating nonpregnant cows, in early as well as late lactation. We expected that among dairy cows kept under similar feeding conditions, the low compared with high CH_4 yielding cows are characterized by (1) a lower innate and adaptive immune response, (2) a similar FCE and similar or greater MPE, and (3) a difference in the community structure of methanogenic archaea (e.g., *Methanosphaera*) and carbohydrate-degrading bacteria.

MATERIALS AND METHODS

The experimental protocol complied with the German Animal Welfare Act guidelines and was approved by the ethics committee of the State Government in Mecklenburg-Western Pomerania (Registration no. 7221.3–1-015/19).

Experimental Design

The data sampling was part of a larger research project conducted between January and December 2020. Two datasets were generated in parallel, one with 20 early (32 ± 7 DIM) and one with 14 late (359 ± 90 DIM) lactating Holstein dairy cows in their second to fourth lactation, which were randomly selected from the herd of the Research Institute for Farm Animal Biology (Dummerstorf, Germany). All cows were clinically healthy, not pregnant, and kept in a freestall barn under comparable climate, housing, and management conditions. All cows were well adapted to the respiration chamber (described in detail in the next section) before entering a 5-d experimental protocol. On d 1 of the experiment, cows were transferred from the freestall barn into one of the 4 open-circuit respiration chambers. After an overnight stay to allow equilibration of gas concentrations, CH_4 production, feed intake, and milk yield were measured for 24 h. On d 4, cows were transported from the respiration chamber back into freestall barn, in which on d 5, a blood and a ruminal fluid sample was taken. Because blood could not be collected in the chamber, it was withdrawn 24 h after transport to the freestall barn to minimize the effect of transportation on blood parameters. According to daily CH_4 production and DMI measured in respiration chamber, cows were retrospectively divided into an equal number of cows with low (**LMY**) and high (**HMY**) CH_4 yield categories within each lactation group. In early lactation, LMY cows (n = 10) had a CH_4 yield of (LSM) 18.7 g/kg of DMI, and HMY cows (n = 10) yielded 25.3 g/kg of DMI. In late lactation, LMY cows (n = 7) averaged had a CH_4 yield of 22.8 g/kg of DMI, and HMY cows (n = 7) yielded 26.8 g/kg of DMI. The lactation numbers in LMY and HMY cows were both second to third in early lactation, and second to third and second to fourth in late lactation, respectively. The DIM in LMY and HMY cows were 27 ± 2 and 35 ± 9 in early lactation, and 374 ± 107 and 345 ± 75 in late lactation, respectively.

Milking and Feeding Management in the Freestall Barn

In the freestall barn, cows were milked twice daily at 0500 h and 1630 h, had ad libitum access to drinking water, and were fed ad libitum a TMR twice daily at 0430 h and 1530 h. The diet was formulated to meet the energy and nutrient requirements according to the recommendations of the German Society of Nutrition Physiology (GfE, 2004). The forage-to-concentrate ratio was, on a DM basis, 67:33 and 65:35, for early- and late lactating cows, respectively. The proportions of forage (grass and corn silage, barley straw) and concentrate constituents differed between lactation stages (Table 1). Overall, the CP, crude fat, and ME were 5.5%, 8.8%, and 5.2%, respectively, greater in the diet of early than late lactating cows (Table 1). Early lactating cows received the lactation diet from the day after calving. At 5 DIM cows were transferred from the calving box to the freestall barn. Late lactating cows received the diet at least 3 wk before entering the respiration chambers.

Measurements in Respiration Chambers

Before the actual measurement of individual CH_4 production, cows were trained and adapted to the respiration chambers 3 to 4 times with increasing time of stay (4–6, 8–10, and 24–26 h) to ensure normal feeding, drinking, and ruminating behavior. Early lactating cows were adapted to the respiration chamber in the preceding lactation, whereas late lactating animals were adapted during midlactation. On d 1 and before entering the respiration chamber, the BW of each cow was determined on an electronic scale, and the metabolic BW (**$BW^{0.75}$**) was calculated. Twenty early lactating cows were measured on 16 calendar days, thereof 13 cows on 13 different calendar days, 2 pairs of cows on a 14th and 15th calendar day, and 3 cows on a 16th calendar day. All animals were part of a larger study. Fourteen late lactating cows were measured in groups of 2 to 4 animals on 7 calendar days. Each chamber had a dimension of 4 × 2 × 2 m containing a 2.5 × 1.5 m tiestall that allowed the cow to voluntarily stand or lie down. Ambient air was drawn through the chamber by a rotary vane vacuum pump (VT 4.40, Fürgut GmbH, Tannheim, Germany), and the temperature was kept at 15°C with a dark-light cycle from 0600 to 1900 h. The airflow through the chamber was approximately 30 m^3/h, which was measured by a differential-pressure type V cone flow meter (McCrometer, Hemet, CA). The CH_4 and CO_2 concentrations were analyzed by infrared absorption, and the O_2 concentration was analyzed paramagnetically (SIDOR, Sick AG, Waldkirch, Germany) at 6-min intervals (Derno et al., 2013). The mean recovery rate from the 4 respiration chambers was 99.7% and determined by CH_4 injections, as described previously by Mesgaran et al. (2020). On d 2, gas exchange measurement began at 0700 h and lasted for the next 24 h. The daily mean of the measurement period lasting from 0700 to 0659 h of the next day was calculated. For milking and cleaning procedures, staff entered the chamber through an air-lock rinsed with chamber air, which prevented changes in gas concentrations. In the chamber, each cow was tied, had ad libitum access to water, and their feed intake was measured by feed disappearance from a feeding trough connected to an electronic scale. Animals were fed twice daily at 0700 and 1600 h. On d 2, TMR samples were collected, pooled (n = 2), and stored frozen at −20°C. Cows were milked at 0715 and 1630 h.

Table 1. Ingredients and chemical composition of feed offered to cows in early and late lactation

Item	Early lactation		Late lactation	
	Mean	SD	Mean	SD
Ingredient (g/kg of DM)				
Grass silage[1]	303.9	62.32	344.1	68.07
Corn silage	339.1	37.26	281.5	27.56
Concentrate[2]	195.4	14.47	166.6	2.35
Barley straw	29.6	8.90	27.5	—
Corn meal	43.4	15.37	48.0	5.29
Wheat seeds	14.1	2.86	57.5	53.77
Rapeseed extraction meal	43.8	9.14	50.1	29.40
Soybean meal	18.2	8.67	12.9	0.18
Soya oil	0.9	0.27	0.6	0.01
Mineral feed[3,4]	5.7	0.96	6.2	1.60
Limestone[5]	2.5	1.04	3.1	0.36
Nutrients (g/kg of DM)				
Crude ash	71.4	16.48	66.4	10.84
Crude protein	161.3	36.30	152.1	12.31
Crude fiber	143.0	43.99	157.1	6.62
Crude fat	32.9	1.05	30.0	3.88
Sugar	35.9	11.05	22.0	9.75
Starch	227.0	60.76	253.4	13.47
ME, MJ/kg of DM	11.5	2.58	10.9	0.72

[1]Combination of grass silage, rye grass silage, and grass-alfalfa silage.

[2]MF2000 (Vollkraft Mischfutterwerke GmbH, Güstrow, Germany): 33% extracted soy meal, 20% corn, 17% wheat gluten, 13% wheat, 8% extracted rapeseed meal, 5% sugar beet pulp, 2% sodium hydrogen carbonate, 1.3% calcium carbonate, and 0.2% sodium chloride and beet pulp-concentrate.

[3]Panto Mineral R 8609 (HL Hamburger Leistungsfutter GmbH, Hamburg, Germany): 20% calcium, 6% phosphorus, 8% sodium, 6% magnesium, 0.03% inorganic nitrogen, 13.74% phosphorus pentoxide. Additives per kilogram original substance: 900,000 IU vitamin A, 200,000 IU vitamin D_3, 4.5 g vitamin E, 1.5 g Cu, 8 g Zn, 5 g Mn, 60 mg I, 21 mg Co, 50 mg Se.

[4]Animal Feed Salt (Esco – European salt company GmbH & Co. KG, Hannover, Germany): 38% sodium, 0.3% calcium, 0.01% magnesium.

[5]Bergophor $CaCO_3$ V001 (Hohburg Mineralfutter GmbH, Lossatal, Germany): 37% calcium.

Milk samples from the preceding evening and morning milking were proportionally pooled and conserved with bronopol. The target volume of the milk sample pool was 100 mL diluted with 130 μL of bronopol. Samples from the evening milking were stored at 4°C overnight before pooling with the morning milk sample. Pooled samples were delivered the same day of pooling at ambient temperature to the Landeskontrollverband für Leistungs- und Qualitätsprüfung Mecklenburg-Vorpommern e.V. (Güstrow, Germany) for analysis.

Feed and Milk Analyses and Calculations

The DM content of the feed samples was determined by drying first at 60°C for 24 h and then at 103°C for 4 h. Samples were ground through a 0.7-mm round-hole screen with a rotary cutting mill (Brabender, Duisburg, Germany). Analysis of the feed was performed at the Landwirtschaftliche Untersuchungs- und Forschungsanstalt (LUFA) in Rostock, Germany. Samples were ashed in a muffle furnace at 550°C for 2 h to obtain crude ash (**CA**). Nitrogen (N) concentration was determined by the Dumas method using an Elementar rapid N III Analyzer (Elementar Analysensysteme, Hanau, Germany) to allow the calculation of CP concentration as 6.25 × N. Crude fiber (**CF**) was determined after treatment of feed samples with HCl and NaOH, and crude fat (**CL**) after the extraction with petrol ether. Starch was determined by polarimetry and N-free extracts (**NfE**) containing sugar, hemicellulose and pectins as follows: NfE = 100 − water − CA − CP − CL − CF (Naumann et al., 2007). The ME (MJ/kg DM of the diet) was calculated as previously described by Bielak et al. (2016):

$$\mathrm{ME} = 6.0756 + 0.19123\ \mathrm{CL} + 0.02459\ \mathrm{CP} - 0.000038\ \mathrm{CF}^2 - 0.002139\ \mathrm{CL}^2 - 0.00006\ \mathrm{CP}^2.$$

Milk composition was analyzed for milk fat, protein, and lactose to calculate ECM yield according to Sjaunja et al. (1990) as follows:

$$\mathrm{ECM\ (kg)} = \text{milk yield (kg/d)} \times [0.038 \text{ x milk fat (g)} + 0.024 \times \text{milk protein (g)} + 0.017 \times \text{milk lactose (g)}]/3.14.$$

The FCE was calculated as a ratio of ECM yield (kg) to DMI (kg). The MPE was determined as ECM yield (kg) per 100 kg of BW and as ECM yield (g) per kilogram of $BW^{0.75}$. The CH_4 emissions were expressed as follows: (1) absolute daily CH_4 production (g/d); (2) CH_4 yield, defined as g of CH_4/kg of DMI; and (3) CH_4 intensity, defined as g of CH_4/kg of ECM yield.

Blood Sampling and Analyses

Sampling. Blood samples were collected by either puncture or catheterization of the jugular vein in Vacuette blood collection tubes containing sodium heparin (Greiner Bio-One International, Frickenhausen, Germany).

Whole Blood Assay. Heparinized blood was diluted 5-fold (200 μL blood and 800 μL medium) in RPMI-1640 medium (PAN-Biotech, Aidenbach, Germany) supplemented with 10% fetal bovine serum, 2 m*M* L-glutamine (PAN-Biotech), 50 μg/mL gentamycin (Sigma-Aldrich/Merck, Taufkirchen, Germany) and 0.05 m*M* mercaptoethanol (Sigma-Aldrich/Merck), and stimulated with a final concentration of 12.5 μg/mL LPS and without LPS (control, supplemented medium only; Sigma-Aldrich/Merck). After a 24-h incubation at 37°C and 5% CO_2, 24-well plates were centrifuged at 220 × *g* for 10 min at 20°C and supernatants were collected and stored at −72°C until analyses. Cow tumor necrosis factor (TNF) ELISA kit (abx150208; Hölzel Diagnostika Handels GmbH, Köln, Germany) was used according to Bacou et al. (2017) and the manufacturer's instruction to quantify TNF-α with a detection limit of 3.7 pg/mL. Supernatants were analyzed in duplicates on the same plate and the CV between optical densities (**OD**) of the duplicates were calculated. If the coefficient of variation was >0.1, the analysis of this sample was repeated.

Proliferation Assay of PBMC. The PBMC were isolated by density-gradient centrifugation. The erythrocytes were lysed (1 mL H_2O for 15 s), osmolarity reinstalled with 108 μL of NaCl solution (8.8%). The PBMC were resuspended with RPMI-1640 medium supplemented as described previously (Meese et al., 2018). Cell number and viability were determined using an automatic cell counter (Multisizer 3 Coulter Counter, Beckman Coulter, Krefeld, Germany). Isolated cell suspensions were seeded onto 96-well microplates in triplicate per treatment after adjusting the cell concentration to 2×10^6 cells/mL. The PBMC were incubated in humidified air with 5% CO_2 at 37°C for 24 h. T-cell proliferation in the presence or absence of the mitogens phytohemagglutinin L (**PHA**; 4 μg/mL) or concanavalin A (**ConA**; 6.25 μg/mL) was assessed using 3-[4,5-dimethyldiazol-2-yl]-2,5 diphenyl tetrazolium bromide (**MTT**; Sigma-Aldrich/Merck) as described by (Meese et al., 2018). After 72 h, cells were treated with MTT and incubated for 4 h and then overnight after application of SDS (Sigma-Aldrich/Merck). The OD was measured with the same microplate reader with test and reference wavelengths of 550 and 690 nm, respectively. The proliferation index (**PI**) was calculated as

$$\mathrm{PI} = \mathrm{OD}_{\text{presence of mitogen}}/\mathrm{OD}_{\text{absence of mitogen}}.$$

Rumen Fluid Sampling and Analysis. Rumen fluid samples were collected via esophageal tubing connected to a vacuum pump as described previously (Muizelaar et al., 2020). The first approximately 100 mL was discarded to eliminate saliva contamination. The subsequent 200 mL of fluid was collected and a subsample of 2 mL from each cow was snap frozen in liquid nitrogen and stored at −80°C before microbial analysis.

Microbial Community Analysis. The DNA extraction, sequencing library preparation, DNA sequencing and bioinformatic processing was performed by DNASense (Aalborg Øst, Denmark) as described in the following sections.

DNA Extraction: The DNA of rumen fluid samples of all 34 cows were extracted using a slightly modified version of the standard protocol for FastDNA Spin kit for Soil (MP Biomedicals) with the following exceptions: 500 µL of sample, 480 µL sodium phosphate buffer and 120 µL MT buffer (lysis enhancer) were added to a Lysing Matrix E tube. Bead beating was performed at 6 m/s for 4 × 40 s (Albertsen et al., 2015). Gel electrophoresis using Tapestation 2200 and Genomic DNA screentapes (Agilent) was used to validate product size and purity of a subset of DNA extracts. DNA concentration was measured using a Qubit dsDNA HS/BR Assay kit (Thermo Fisher Scientific).

Sequencing Library Preparation: Amplicon libraries for the bacteria/archaea 16S rRNA gene variable region 4 were prepared by a custom protocol based on an Illumina protocol (Illumina, 2015). The PCR conditions were set with the following program: initial denaturation at 95°C for 2 min, 30 cycles of amplification (95°C for 15 s, 55°C for 15 s, 72°C for 50 s) and a final extension at 72°C for 5 min. Duplicate PCR reactions were performed for each sample and the duplicates were pooled after PCR. The forward and reverse, tailed primers were designed according to Illumina (2015) and contained primers targeting the bacteria/archaea 16S rRNA gene variable region 4: [515FB] GTGYCAGCMGCCGCGGTAA and [806RB] GGACTACNVGGGTWTCTAAT (Apprill et al., 2015). The resulting amplicon libraries were purified using the standard protocol for CleanNGS SPRI beads (CleanNA, PH Waddinxveen, the Netherlands) with a bead-to-sample ratio of 4:5. DNA concentration was measured using a Qubit dsDNA HS Assay kit (Thermo Fisher Scientific). Gel electrophoresis using Tapestation 2200 and D1000/High sensitivity D1000 screentapes (Agilent) was used to validate product size and purity of a subset of sequencing libraries. Sequencing libraries were prepared from the purified amplicon libraries using a second PCR. This PCR reaction (25 µL) contained PCRBIO HiFi buffer (1×), PCRBIO HiFi Polymerase (1 U/reaction; PCRBiosystems Ltd., London, UK), adaptor mix (400 n*M* of each forward and reverse), and up to 10 ng of amplicon library template. The PCR was set to the following program: initial denaturation at 95°C for 2 min, 8 cycles of amplification (95°C for 20 s, 55°C for 30 s, 72°C for 60 s) and a final elongation at 72°C for 5 min.

DNA Sequencing: The purified sequencing libraries were pooled in equimolar concentrations and diluted to 2 n*M*. The samples were paired end sequenced (2 × 300 bp) on a MiSeq (Illumina) using a MiSeq Reagent kit v3 (Illumina) following the standard guidelines for preparing and loading samples on the MiSeq. Sequence files associated with each sample can be accessed at the European Nucleotide Archive (https://www.ebi.ac.uk/ena/browser/home), accession number PRJEB85754.

Bioinformatic Processing: Forward and reverse reads were trimmed for quality assessment using Trimmomatic v.0.32 (Bolger et al., 2014) with the settings SLIDINGWINDOW:5:3 and MINLEN: 225. The trimmed forward and reverse reads were merged using FLASH v.1.2.7 (Magoč and Salzberg, 2011) with the settings -m 10 -M 250. The trimmed reads were dereplicated and formatted for use in the UPARSE workflow (Edgar, 2013). The dereplicated reads were clustered, using the usearch v.7.0.1090 -cluster_otus command with default settings. Operational taxonomic unit (**OTU**) abundances were estimated using the usearch v. 7.0.1090 -usearch_global command with -id 0.97 -maxaccepts 0 -maxrejects 0. Taxonomy was assigned to the aligned sequence using the uclust classifier as implemented in the assign_taxonomy.py script in QIIME (Caporaso et al., 2010) and the SILVA database, release 132 (Quast et al., 2013). After initial processing of DNA extracted from the samples, low-quality filtered reads (filtReads less than 10,000) were removed. In the current study an average of 47,307 filtered DNA reads were obtained, with the filtered DNA reads per sample ranging from 26,348 to 72,365, thus all samples were included in the subsequent analysis.

Statistical Analyses

Animal and microbiome data were analyzed using R (version 4.4.1; https://www.r-project.org). Statistical analysis was performed separately for cows in early and late lactation using a linear mixed effects model. The statistical model for analyzing animal data included DMI, milk yield, ECM, BW, FCE, MPE, CH_4 emission, and immune response with CH_4 category (1, 2) and period (date of chamber measurement) as fixed effects and individual animal as random effect. For animal data, estimated marginal mean and SEM were reported using the emmeans package in R (Lenth, 2017). Significant differences were declared at $P < 0.05$, and trends at $0.05 \le P < 0.10$. The period effect was not significant for all variables, except ECM.

For microbiome data, relative abundance of less than 0.1% across all samples in early and late lactation was filtered before performing differential abundance analysis. Within- and between-sample differences in microbial communities were assessed using α- and β-diversities, respectively. Alpha diversity indices, including richness, Shannon, and Simpson, were calculated, and statistical analysis was performed using a linear mixed effect model by including CH_4 category as a fixed effect and individual cow as a random effect. Beta diversity was calculated using Bray–Curtis dissimilarity and visualized with principal coordinate analysis (**PCoA**) plot. Statistical analysis for β-diversity was performed using the permutational multivariate ANOVA (**PERMANOVA**) test. Significant differences in α- and β-diversity were declared at $P < 0.05$, and tendency at $0.05 \leq P < 0.10$. Linear discriminant analysis effect size (**LEfSe**) was performed to test differences in 16S rRNA bacterial and archaeal amplicon between LMY and HMY cows within lactation group, and significant differences were identified by the combined criteria of Kruskal–Wallis $P < 0.05$, Wilcoxon P-adjusted <0.05, and linear discriminant analysis (**LDA**) score >3.

RESULTS

Feed Intake, Performance, and Gaseous Exchange in Low and High CH_4 Yielders

Intake, performance, and gas exchange data are shown in Table 2. In early lactating animals, the BW of LMY and HMY ranged from 545 kg to 779 kg and was not different between CH_4 categories ($P = 0.37$). Dry matter intake was 1.4-fold higher ($P < 0.01$) in LMY compared with HMY cows. We observed no differences ($P > 0.15$) in milk yield and composition between LMY and HMY cows. Accordingly, the FCE was lower ($P = 0.01$) in LMY than HMY, whereas MPE (kg of ECM/100 kg of BW and kg of ECM/$BW^{0.75}$) was greater ($P = 0.03$) in LMY compared with HMY cows. The daily CH_4 production (g/d) did not differ ($P = 0.27$) between LMY and HMY cows. The CH_4 yield (g/kg of DMI) was 25.4% lower ($P < 0.01$) in the LMY compared with the HMY category, whereas CH_4 intensity expressed as grams/100 kg of BW, grams per kilogram of $BW^{0.75}$, and grams per kilogram of ECM did not differ ($P > 0.10$) between LMY and HMY animals. Daily CO_2 production did not differ ($P = 0.11$) between LMY and HMY cows. In LMY animals, CO_2 yield (g/kg of DMI) was lower ($P < 0.01$), but CO_2 per 100 kg BW and per kilogram of $BW^{0.75}$ were greater ($P = 0.02$) than in HMY cows. The CH_4:CO_2 ratio was similar ($P = 0.79$) between CH_4 categories in the current study. In late lactating cows, the animal categories did not differ ($P \geq 0.12$) in any of the above-mentioned traits on animal performance and gas emissions except CH_4 yield, with HMY cows producing 16.1% more ($P < 0.01$) than LMY animals, and CH_4 intensity (g/kg of ECM), which tended to be greater ($P = 0.08$) in HMY compared with LMY animals.

In Vitro Response of Blood Cells

In early lactating cows, LMY compared with HMY animals tended to have a lower PI of PBMC in response to PHA ($P = 0.051$) and ConA ($P = 0.050$), and a lower ($P = 0.02$) TNF-α response of whole blood to LPS stimulation (Figure 1A–C, respectively). Additionally, the OD of unstimulated PBMC in LMY were greater ($P = 0.04$) compared with HMY in early lactation, whereas OD of stimulated PBMC to ConA and to PHA and TNF-α concentration of unstimulated whole blood did not differ ($P \geq 0.41$) in HMY compared with LMY animals (Supplemental Table S1, see Notes). For late lactating cows, no differences ($P \geq 0.23$) were observed in immune response variables (Figure 2A–C). No differences ($P \geq 0.15$) occurred in the OD of unstimulated PBMC or the OD of stimulated PBMC to PHA and TNF-α concentration of unstimulated whole blood between LMY and HMY, but the OD of stimulated PBMC to ConA was higher ($P = 0.02$) in LMY than HMY (Supplemental Table S1).

Rumen Microbial Community Structure

A total of 2,918 OTU that could be taxonomically classified to genus level were identified, corresponding to 264 distinct genera of bacteria and archaea across all samples. Overall, the bacterial community at the phylum level was dominated by *Firmicutes* (mean; min-max; 56%; 32%–73%) and *Bacteroidetes* (33%; 14%–49%; Supplemental Figure S1A, C, D, E, see Notes). *Actinobacteria* was detected at low abundance (4%; 0.7%–19%). Among all identified bacterial genera, *Prevotella* 1 was found to be the most abundant across all samples, accounting for 19% (7%–31%). Other bacterial genera, including *Succiniclasticum* (7.8%; 1.7%–13%), *Christensenellaceae* R-7 group (7.2%; 2.1%–15%), *Lachnospiraceae* NK3A20 (6.5%; 1.1%–14%), *Ruminococcus* 2 (4.8%; 0.3%–11%), *Ruminococcaceae* NK4A214 group (4.6%; 1.4%–11%), and *Acetitomaculum* (3.1%; 0.4%–7.9%), were detected at low abundance (Supplemental Figure S1B). For the archaeal community, only 4 genera were detected, with *Methanobrevibacter* accounting for the most abundant genera (91%; 82%–92%), followed by *Methanosphaera* (8%; 4%–18%). The remaining genera were rare, including *Candidatus*, *Methanomethylophillus*, unclassified *Methanomethillophilaceae*, unclassified *Nitrososphaeraceae*, and *Methanimicrococcus* (Supplemental Figure S1F).

Rumen Microbiome Structure in Low Versus High CH_4 Yielders

In early lactating cows, the 4 analyzed matrixes of α diversity, including constituted number (observed OTU

Table 2. Performance and gas emissions of cows with low (LMY) and high CH_4 yield (HMY)[1] in early and late lactation

Item	Early lactation LMY	Early lactation HMY	SEM	*P*-value	Late lactation LMY	Late lactation HMY	SEM	*P*-value
BW (kg)	638.4	670.4	19.7	0.37	810.9	768.6	25.4	0.70
DMI (kg/d)	20.1	14.0	1.0	<0.01	18.1	15.0	1.1	0.08
Milk production and composition								
Milk yield (kg/d)	41.1	35.8	1.79	0.22	20.1	17.6	1.36	0.15
ECM yield (kg/d)	44.3	38.1	1.63	0.10	22.4	20.7	1.20	0.25
Milk fat (%)	4.6	4.3	0.16	0.45	4.6	5.0	0.40	0.34
Milk protein (%)	3.2	3.3	0.10	0.59	4.0	3.9	0.13	0.96
Milk lactose (%)	5.0	5.1	0.06	0.41	4.7	4.7	0.07	0.35
Milk fat (kg/d)	1.9	1.5	0.08	0.06	0.9	0.9	0.06	0.81
Milk protein (kg/d)	1.3	1.2	0.06	0.38	0.8	0.7	0.06	0.17
Milk lactose (kg/d)	2.1	1.8	0.09	0.32	1.0	0.8	0.07	0.15
Milk fat:protein	1.5	1.3	0.07	0.36	1.2	1.3	0.12	0.38
lnSCC[2]	1.4	1.4	0.09	0.71	2.1	2.2	0.09	0.24
Efficiency[3]								
FCE (ECM/DMI, kg/kg)	2.3	2.8	0.19	0.01	1.2	1.4	0.13	0.26
MPE (ECM/BW, kg/100 kg)	7.0	5.7	0.27	0.03	2.8	2.7	0.17	0.43
MPE (ECM/BW, g/kg$^{0.75}$)	349.7	289.1	12.5	0.03	149.1	142.7	8.64	0.37
Gas emissions								
CH_4 (g/d)	376.8	350.5	20.2	0.27	415.3	399.0	27.2	0.91
CH_4/DMI (g/kg)	18.7	25.3	0.88	<0.01	22.8	26.8	0.87	<0.01
CH_4/BW (g/100 kg)	59.4	52.2	3.09	0.10	51.6	51.8	3.76	0.87
CH_4/BW (g/kg$^{0.75}$)	3.0	2.7	0.15	0.12	2.7	2.7	0.19	0.88
CH_4/ECM (g/kg)	8.5	9.5	0.60	0.79	18.9	19.7	0.91	0.08
CO_2 (g/d)	13,317	11,883	558	0.11	14,201	13,116	553.4	0.51
CO_2/DMI (g/kg)	668.2	865.7	35.32	<0.01	784.3	897.1	77.35	0.20
CO_2/BW (g/100 kg)	2,102	1,768	80.2	0.02	1,763	1,703	60.6	0.61
CO_2/BW (g/kg$^{0.75}$)	105.4	90.0	3.93	0.02	93.9	89.6	3.11	0.54
CO_2/ECM (g/kg)	301.1	320.8	16.28	0.96	650.9	647.1	25.39	0.28
CH_4:CO_2 (g/kg)	28.3	29.4	1.0	0.79	29.1	30.5	1.9	0.55

[1]Cows with low (LMY) and high CH_4 yield (HMY) were determined retrospectively and within each lactation group separately based on daily CH_4 production and DMI assessed in respiration chambers.

[2]lnSCC = natural logarithm of SCC ($\times 10^3$ cells/mL). The SCC ($\times 10^3$ cells/mL) of LMY versus HMY in early lactation was 21.2 versus 23.7, and in late lactation 185 versus 193.

[3]FCE = feed conversion efficiency, defined as kilograms ECM/kilograms of DMI; MPE = milk production efficiency, defined as either kilograms ECM/100 kilograms of BW or kilograms ECM/kilograms of BW$^{0.75}$.

and richness) and distribution (Shannon and Simpson indexes), did not differ ($P \geq 0.20$) between LMY and HMY categories. In late lactation animals, the 3 α diversity matrixes, OTU, richness, and Simpson, did not differ ($P \geq 0.20$) between CH_4 categories, whereas Shannon α diversity tended to be greater ($P = 0.06$) in HMY than LMY animals (Supplemental Table S3, see Notes). In early and late lactation, the rumen content of LMY and HMY cows did not differ (PERMANOVA test, $P \geq 0.22$) for β diversity (PCoA plot; Figure 1D, Figure 2D). In early lactation, LEfSe analysis showed archaea in the genus *Methanosphaera* and bacteria in the genus *Marvynbryantia* were significantly enriched in the rumen content of LMY animals, and the relative abundance of *Butyrivibrio 2* was found to be elevated in HMY cows (Figure 1E). In late lactation, LEfSe analysis revealed bacteria in the genus *Prevotella 7*, *Ruminococcus gauvreauii group*, *Eubacterium ruminantium group*, and *Shuttleworthia* were enriched in rumen content of LMY animals, whereas in HMY cows, multiple taxa were enriched, including, but not limited to, archaeal and bacterial genera *Methanobrevibacter*, *Succinivibrionaceae UCG-002*, *Rikenellaceae RC9 gut group*, *CAG-352*, and *Veillonellaceae UCG-001* (Figure 2E).

DISCUSSION

The statistical comparison between lactation groups could not be done in our study because early and late lactating cows were different individuals, fed different diets, and grouped by different cut-offs for defining LMY and HMY (outlined in this discussion), therefore any conclusions drawn are limited to the respective lactation.

Methane Categorization in Early and Late Lactation

In our study, the categories of low and high CH_4 yield were determined retrospectively by dividing cows into 2 equal numbers of animals based on their CH_4 yield values, resulting in average CH_4 yields of 18.7 versus

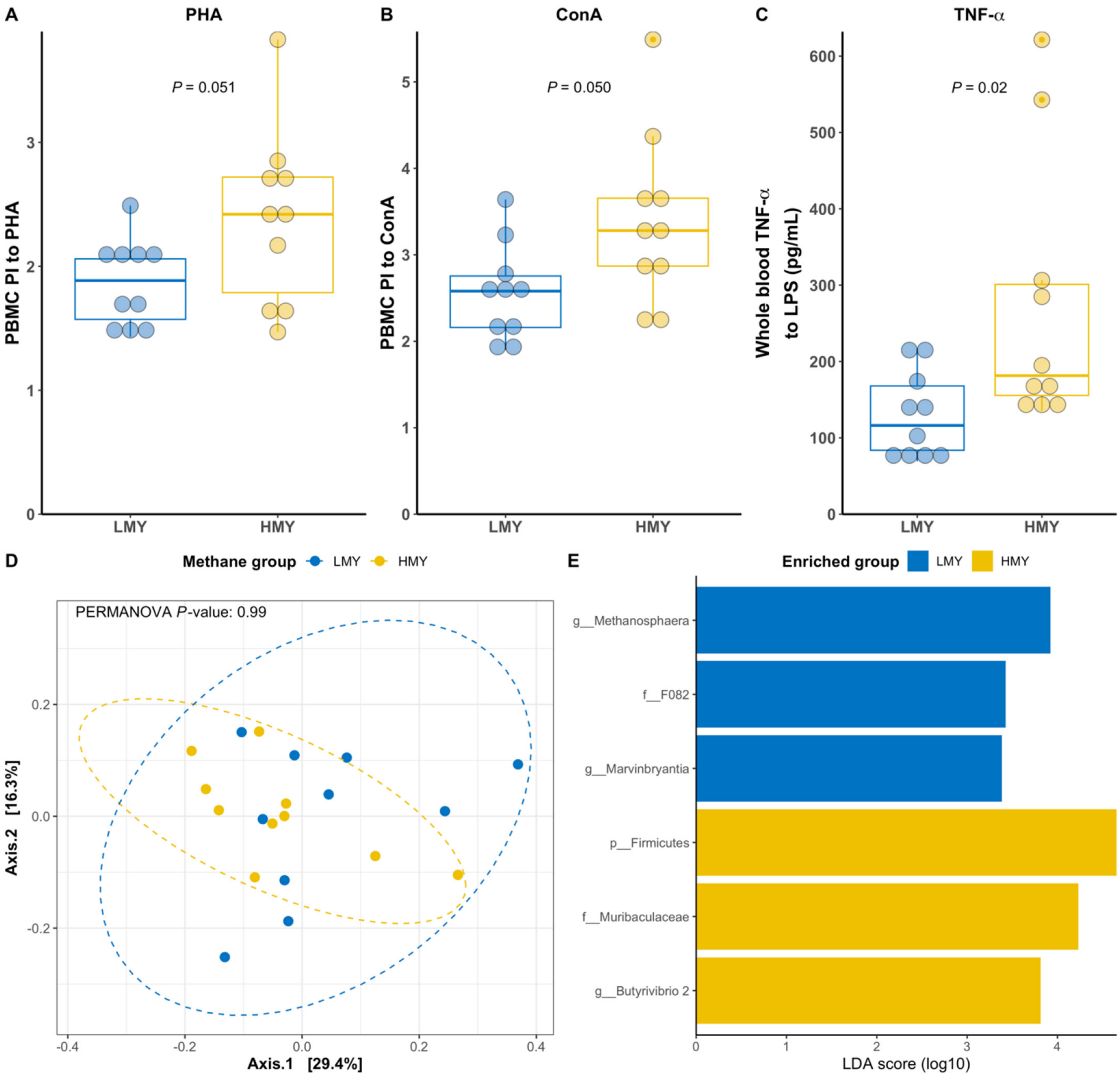

Figure 1. Early lactating cows with low (LMY) and high (HMY) CH_4 yield: Proliferation index (PI) of peripheral blood mononuclear cells (PBMC) in response to phytohemagglutinin (PHA; A) and concanavalin A (ConA; B), whole blood tumor necrosis factor (TNF)-α response to LPS (C), β diversity indices using principal coordinate analysis (PCoA) plots with Bray–Curtis distance method (D), and histogram illustrating significantly different microbial taxa in LEfSe analysis (E). Taxa in panel E are labeled with their taxonomic rank: p_ = phylum, f_ = family, and g_ = genus. In the box plots, boxes represent the interquartile range from the 25th to 75th percentiles, the midline shows the median, the whiskers extend the most extreme values within 1.5× the interquartile range, and dots corresponds to all data points.

25.3 g CH_4/kg of DMI in early lactation (32 DIM) and 22.8 versus 26.8 g CH_4/kg of DMI in late lactation (376 DIM). Our approach was different from previous studies where cows with medium CH_4 yields were removed from the dataset (Lyons et al., 2018; Denninger et al., 2020; Stepanchenko et al., 2023). It must be kept in mind that our approach assigns similar cows in the middle of the distribution curve to either low or high methane category group, potentially washing out the differences between phenotypes. Within the lactation stage we bal-

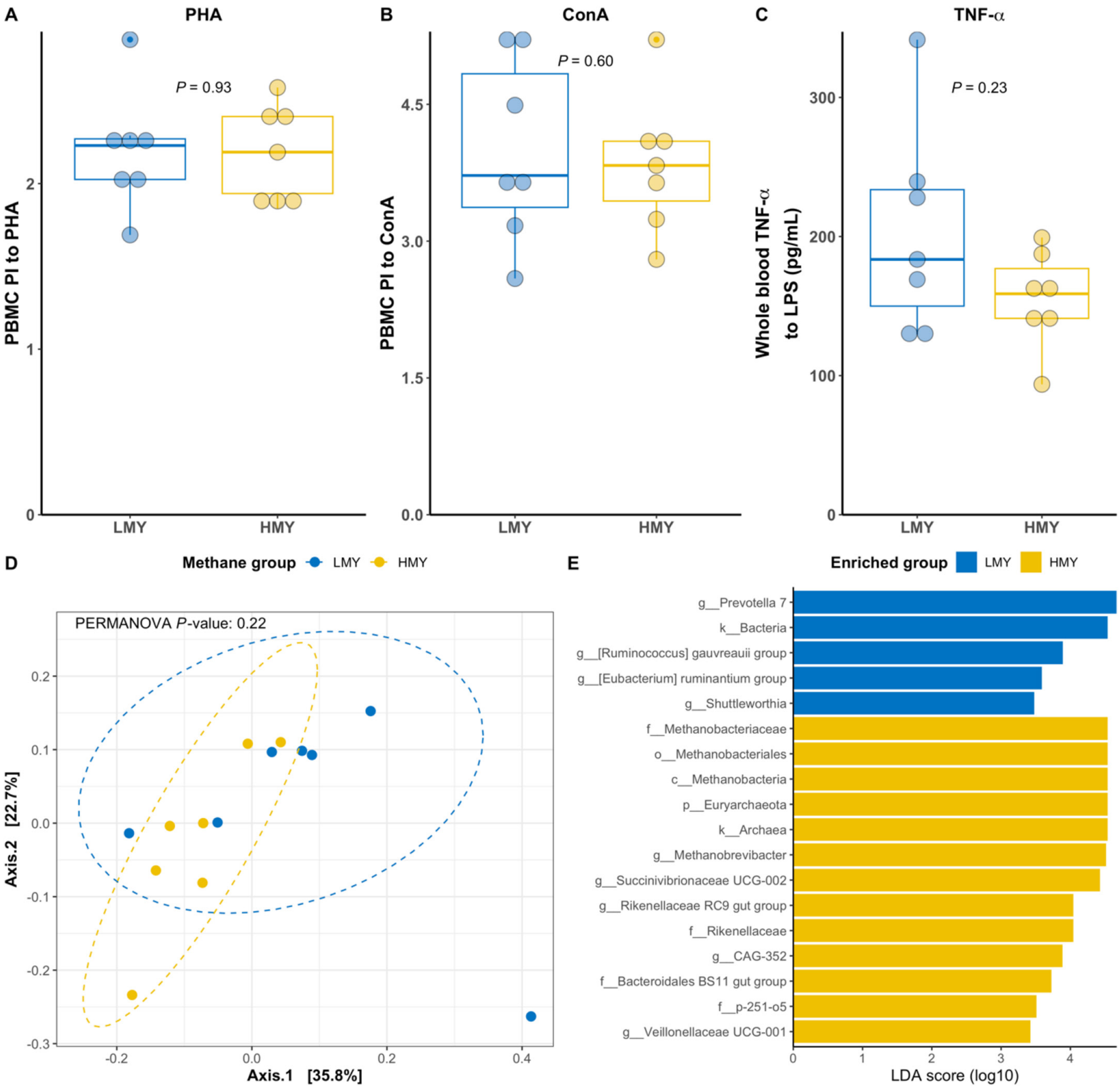

Figure 2. Late lactating cows with low (LMY) and high (HMY) CH_4 yield: Proliferation index (PI) of peripheral blood mononuclear cells (PBMC) in response to phytohemagglutinin (A; PHA) and concanavalin A (B; ConA), whole blood tumor necrosis factor (TNF)-α response to lipopolysaccharide (C; LPS), β diversity indices using principal coordinate analysis (PCoA) plots with Bray–Curtis distance method (D), and histogram illustrating significantly different microbial taxa in LEfSe analysis (E). Taxa in panel E are labeled with their taxonomic rank: k_ = kingdom, p_ = phylum, c_ = class, o_ = order, f_ = family, and g_ = genus. In the box plots, boxes represent the interquartile range from the 25th to 75th percentiles, the midline shows the median, the whiskers extend the most extreme values within 1.5× the interquartile range, and dots corresponds to all data points.

anced the number of animals per methane category to be equal, resulting in different cut-off values defining low and high CH_4 yielding animals, namely 21.5 g CH_4/kg of DMI in early lactation and 24.5 g CH_4/kg of DMI in late lactation. To the best of our knowledge there is no reported cut-off value defining low and high CH_4 yielding animals. For instance, in the study by Stepanchenko et al. (2023) on late lactation Holstein cows (250 DIM), the low versus high CH_4 yielders produced on average of 13.5 versus 17.7 g CH_4/kg of DMI, respectively. Overall higher CH_4 yields, more similar to the cows in our study, were reported by Denninger et al. (2020) for late lactating Brown Swiss and Braunvieh dairy cows (241 DIM), where the low versus high CH_4 yielders produced on average 21.7 versus 25.6 g CH_4/kg DMI, respectively. Similarly, the longitudinal study by Lyons et al. (2018) reported on average 20.7 and 27.2 g CH_4/kg DMI in low and high CH_4 yielding late lactating Holstein cows, respectively. In a future studies, more animals should be screened, followed by a more balanced approach for the selection of low and high CH_4 yielding cows, for example, by choosing animals with comparable DMI.

Assessment of Immune Status

In this study, we investigated the innate and adaptive immune response by exposing whole blood and isolated PBMC to a challenge in vitro, similar to Meese et al. (2018, 2020) and Wang et al. (2018, 2019). In whole blood cell culture, a greater LPS-induced TNF-α production represents improved innate immune response and consequently a lower risk for infectious diseases. The PBMC consist predominantly of T and B lymphocytes, whereas the mitogens ConA and PHA primarily stimulate T lymphocytes. It was assumed that the immune response to a mitogen in vitro reflects the response to an antigen in vivo. Accordingly, with comparatively lower proliferation of PBMC, the risk of infectious diseases should be higher (Catalani et al., 2013). Previous studies reporting CH_4 yield and evaluating immune cell status or response in cattle are scarce and used similar (e.g., Meese et al., 2020) or different methods such as SCC in milk (Arndt et al., 2015). Thus, it must be kept in mind that studies are not completely comparable when it comes to discussing the immune status, and therefore the results reported herein need to be interpreted within the context of the particular methodology used.

Immune Response, Feed Efficiency, and Rumen Microbiome in Cows Differing in CH_4 Yield in Early Lactation

As expected, blood from LMY cows showed a lower innate and adaptive immune response in vitro compared with blood from HMY cows, which was interpreted as LMY cows being more susceptible to infectious diseases in early lactation. Meese et al. (2020) found low-immune-responder Holstein cows in early lactation (11 DIM) to be characterized by a low CH_4 intensity (g CH_4/kg of ECM). The current study used the same immune assay as Meese et al. (2020)—the in vitro response of PBMC to PHA, with an additional mitogen (ConA)—and in addition the whole blood assay confirming the findings of Meese et al. (2020). A decreased immune response in early lactation can be a resource-driven response to the energy limitation and prioritization of the mammary gland leading to an increase in the number and severity of metabolic and infectious diseases that are often observed during this time (Goff, 2006). In the current study, early lactating LMY and HMY cows did not differ in daily CH_4 production, but LMY cows had greater DMI than HMY cows, explaining the CH_4 yield category. Obviously, with greater feed intake after calving LMY cows may achieve a positive EB more quickly and lose relatively less feed gross energy as CH_4, which is supported by the estimated EB in LMY compared with HMY cows (being positive or less negative in LMY than HMY cows, depending on the equation used; Supplemental Table S2, see Notes). Still, for some unknown reason, the "surplus" in energy was not partitioned to enhance milk production, evidenced by similar milk yields of LMY and HMY cows, or to support the blood immune response. One might assume that LMY cows with comparatively higher DMI and similar ECM yield compared with HMY cows would have higher passage rates of digesta. Although passage rate was not measured in the present study, other studies showed that high DMI is linked to an increased passage rate of digesta, leading to a lower ruminal fermentation, which eventually is associated with reduced CH_4 yield (Janssen, 2010; Goopy et al., 2014; Denninger et al., 2020). Additionally, Stepanchenko et al. (2023) found LMY cows to be characterized by a lower fiber digestibility and speculated this result to be due to increased passage rate of digesta. In the present study, however, nutrient digestibility was not assessed, and the observed difference in microbial community was also small (further outlined later in this discussion). It can be assumed that the energy derived from rumen fermentation in LMY cows was not higher than that in HMY cows to the extent expected from the 1.4-fold greater DMI. This was supported by the CO_2 production per unit of DMI, which was 25% lower in LMY than HMY cows and the estimate of CO_2 production deriving from rumen fermentation, which did not differ between LMY and HMY cows (Supplemental Table S2).

During a metabolically challenging time, such as early lactation, blood metabolites from body reserve mobilization can suppress the responsiveness of immune cells (Targowski and Klucinski, 1983). A study by Bielak et al. (2016) found a temporarily lower CH_4 yield (g CH_4/kg of DMI) in high fat mobilizing Holstein cows, but immune parameters were not reported. Nevertheless, due to the comparable milk fat content and fat yield, it is unlikely that our LMY cows had a greater body fat mobilization that could explain the lower immune response.

In the current study, the PI of PBMC in LMY cows was lower because of a comparatively higher basal MTT-reducing activity (OD control, Supplemental Table S1), which is in line with the observation by Meese et al. (2020). The MTT assay does not directly measure proliferation; it measures cellular metabolic activity, primarily mitochondrial activity. Hence, it is possible that LMY cows, along with their immune cells, have a higher basal metabolic rate, which is supported by the higher estimated CO_2 production from animal metabolism in LMY cows (Supplemental Table S2). A higher basal metabolic rate could be at the expense of an adequate immune response and milk production corresponding to feed intake. This is in accordance with the divergent FCE observed in our study. In the present study, LMY cows in early lactation showed a lower FCE than HMY animals, which was unexpected. To the best of our knowledge, no reports are available on the association of CH_4 yield with feed and MPE in early lactation cows. The MPE was greater in LMY as compared with HMY cows, attributed to the numerically lower BW and higher ECM. Our findings are in contrast with reports by Denninger et al. (2020) and Stepanchenko et al. (2023). One explanation for the deviating result is the lactation stage; the reports by Denninger et al. (2020) and Stepanchenko et al. (2023) were on late lactating dairy cows. A study in early lactating dairy cows receiving fibrolytic enzymes at 1.0 mL/kg of TMR showed a greater NDF digestibility in the rumen and an improved FCE (milk yield/DMI) by 10.7% (Holtshausen et al., 2011). Therefore, and in line with the preceding discussion on the lower digestibility and greater basal metabolic rate, the higher DMI but smaller CH_4 energy loss per unit DMI in LMY cows suggest that additional energy loss may have occurred via feces and urine. In contrast, HMY cows may have partitioned sufficient energy toward immune function despite losing more gross energy as CH_4 per unit of feed intake. Nevertheless, it remains valuable to continue studying early lactating cows to understand the variations in CH_4 yield because of differences in daily CH_4 production.

In our study, early lactating LMY and HMY cows did not differ in milk SCC, which averaged 22.5×10^3 cells/mL, well below the 100×10^3 cells/mL threshold (Sharma et al., 2011), indicating the absence of subclinical mastitis and a healthy udder. The SCC is a trait indicative of immune function, with lower SCC indicating increased macrophage killing ability (Gibson et al., 2016). Our result is consistent with studies by Grešáková et al. (2021) and Hailemariam et al. (2021), who found no relationship between CH_4 yield and milk SCC in early, mid, and late lactating Holstein cows. However, our findings are in contrast with Arndt et al. (2015), who reported that low efficient dairy cows (breed not indicated; FCE, milk yield/DMI) in mid-to-late lactation had 3.8-fold higher SCC (107 vs. 488×10^3 cells/mL) and 26% higher CH_4 yield (g CH_4/kg of DMI) compared with highly efficient animals. The discrepancy in FCE and milk SCC results between studies could be explained by the health status of the cows. The absence of subclinical mastitis in our study allowed a higher allocation of ME toward milk synthesis than the immune response (Mezzetti et al., 2024).

Furthermore, the difference in CH_4 yield in early lactation can be partially explained by minor distinction in rumen microbiota between LMY and HMY cows, which was as expected. An elevated abundance of *Methanosphaera*, a methanol-utilizing methanogen, in LMY cows was observed, consistent with Stepanchenko et al. (2023). *Methanosphaera* is typically more prevalent in low CH_4 yielding cows, possibly because more substrates are required to produce one mole of CH_4 via a methylotrophic pathway than via hydrogenotrophic methanogenesis (Pereira et al., 2022). Additionally, *Marvynbryantia*, found to be more abundant in LMY cows, is known for its role in reductive acetogenesis via the Wood–Ljungdahl pathway (Wolin et al., 2003). Reductive acetogenesis can act as an alternative sink to methanogenesis, but under common rumen conditions acetogens are believed to be outcompeted energetically in the rumen by methanogens, except for instances of higher hydrogen partial pressure, when methanogenesis is inhibited (Ni et al., 2025). Although hydrogen was not measured in this study, it can be speculated that a hydrogen increase occurred from our observations of higher *Methanosphaera* abundances in LMY cows (Figure 1E), which follows previous studies that have detected these populations in animals with elevated hydrogen pressure (Pitta et al., 2022). A slight alteration in microbial fermentation arising from intermediates such as hydrogen could additionally indirectly explain why energy for supporting production and immune function may have been depleted in LMY cows in our study, as indicated by their lower FCE (Table 2). In contrast, the observed elevated abundance in HMY cows of bacterial genera *Butyrivibrio*, a fiber-degrading bacteria that produces formate, butyrate, acetate, and H_2 (Van Gylswyk et al., 1996; Emerson and Weimer, 2017; Palevich et al., 2017), may suggest more active rumen fermentation, supporting higher CH_4 production per unit of intake.

Immune Response, Feed Efficiency, and Rumen Microbiome in Cows Differing in CH_4 Yield in Late Lactation

To our knowledge, this is the first study reporting the relationship of CH_4 yield categories with immunological traits, feed efficiency measures, and rumen microbial community in nonpregnant, late lactating dairy cows. The assumed positive EB based on DMI and ECM, and minimum challenges for immune homeostasis in late lactation (Meese et al., 2018; Moreira et al., 2021) and the absence of both metabolic stress and immunosuppression related to pregnancy (Trevisi and Minuti, 2018) allowed us to study the pure effect of the CH_4 yield category.

Unexpectedly, blood cells of LMY cows did not exhibit lower immune response compared with HMY animals. Both innate and adaptive immune responses were similar between CH_4 categories. Previous findings (Meese et al., 2020) and the results of the present study on early lactation cows suggest a positive relationship between enteric CH_4 traits and immune response. But lack of sufficient digestive energy to elicit an adequate immune response in LMY seems to occur only during energy-demanding early lactation. We assume that a more balanced energy status (based on DMI and ECM) may explain why the anticipated immune suppression in LMY cows was not observed in the present study. In the case of pregnant cows in late lactation, it can be speculated that LMY cows may exhibit immunosuppression (Trevisi and Minuti, 2018). Despite the lack of difference in daily CH_4 production and DMI in LMY and HMY cows, the animals differed in CH_4 yield, suggesting a relatively smaller feed energy loss as CH_4 per unit of feed intake in LMY cows. However, consistent with our finding in early lactation, this apparent difference did not appear to translate the energy surplus toward increased milk production, BW gain, and immune function because these parameters were similar between LMY and HMY cows. The overall metabolic activity and energy expenditure did not differ between CH_4 categories based on the similar estimates of CO_2 production deriving from metabolism and heat production in LMY and HMY cows (Supplemental Table S2). Possibly, the surplus of energy from the feed may also be lost in feces and urine to a small extent, which was not measured in this study.

In the present study, similar SCC in milk samples of late lactating LMY and HMY cows were observed. The SCC is an important indicator of udder health, primarily consisting of white blood cells (leukocytes). Therefore, the absence of inflammation in nonpregnant late lactating cows in our study suggests that factors influencing CH_4 yield in Holstein cows do not necessarily affect SCC as previously reported (Grešáková et al., 2021; Hailemariam et al., 2021). The average SCC was 189×10^3 cells/mL, which was relatively high in lactating cows and is consistent with Grešáková et al. (2021).

As expected, the LMY cows were characterized by similar FCE and MPE (ECM/BW), which is consistent with Denninger et al. (2020) and Stepanchenko et al. (2023). This was supported by similar estimates of CO_2 production both from ruminal fermentation and from metabolism (Supplemental Table S2).

As expected, rumen microbiome analysis revealed distinct sets of microbial communities in LMY and HMY animals, and this difference was, interestingly, more pronounced in late than early lactating cows. In LMY cows bacterial genera associated with efficient carbohydrate and fiber degradation were enriched, including *Prevotella* 7, *Ruminococcus gauvreauii* group (Suen et al., 2011; Henderson et al., 2015), and *Shuttleworthia* (Park et al., 2020). In contrast, the rumen content of HMY cows had a greater abundance of the archaeal genus *Methanobrevibacter*, which supports higher CH_4 yield in HMY cows as previously reported (Zhou et al., 2011). Interestingly, *Succinivibrionaceae* UCG-002 was also found to be more abundant in HMY cows, which is in contrast with previous findings that typically associate *Succinivibrionaceae* with low CH_4 yielding animals (Wallace et al., 2015; Stepanchenko et al., 2023). As a succinate producer, *Succinivibrionaceae* has the ability to utilize H_2 and potentially compete with methanogens (Pope et al., 2011). These findings highlight the complexity of rumen microbial interrelation and the challenges of inferring microbial function solely based on relative abundance information. A similar phylogenetic profile in the rumen may not be translated into a similar functional profile, as demonstrated by Stepanchenko et al. (2023). These authors found that although the abundance of methanogenesis-related microbes may have been stable, the difference in gene expression could lead to variation in CH_4 production.

CONCLUSIONS

We demonstrated that in early lactation, low CH_4 yielding cows had lower immune response and FCE with observed minor differences in rumen microbial composition. In late lactating cows, the differences in CH_4 yield were not related to immune response and efficiency measures, but alteration in rumen microbial composition was more pronounced, suggesting a more complex interrelation among microbes. Our study highlights that in early lactation increased CH_4 yield may reflect higher rumen fermentation activity, fostering feed efficiency and energy availability for supporting immune function.

NOTES

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Nonstandard abbreviations used: $BW^{0.75}$ = metabolic BW; c_ = class; CA = crude ash; CF = crude fiber; CL = crude fat; ConA = concanavalin A; EB = energy balance; f_ = family; FCE = feed conversion efficiency; g_ = genus; HMY = high methane yield; k_ = kingdom; LDA = linear discriminant analysis; LEfSe = linear discriminant analysis effect size; LMY = low methane yield; lnSCC = natural logarithmic of SCC; MPE = milk production efficiency; MTT = 3-[4,5-dimethyldiazol-2-yl]-2,5 diphenyl tetrazolium bromide; NfE = N-free extracts; o_ = order; OD = optical density; OTU = operational taxonomic unit; PBMC = peripheral blood mononuclear cells; PCoA = principal coordinates analysis; p_ = phylum; PERMANOVA = permutational multivariate ANOVA; PHA = phytohemagglutinin; PI = proliferation index; TNF-a = tumor necrosis factor-alpha.

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